

ROGER HO

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EDUCATION

CARNEGIE MELLON UNIVERSITY

PhD, Mechanical Engineering

Surgical Mechatronics Lab (Advisor: Cameron N. Riviere)

GPA: 4.00/4.00

Expected 2029

Pittsburgh, PA

UNIVERSITY OF MICHIGAN

BSE Mechanical Engineering; Minor in Computer Science

Overall GPA: 3.92/4.00, summa cum laude | Major GPA: 4.00/4.00

Aug 2019 - May 2023

Ann Arbor, MI

RESEARCH EXPERIENCE

PRECISION SYSTEMS DESIGN LABORATORY

Research Assistant | University of Michigan

May 2022 - May 2023

Ann Arbor, MI

- Devised and conducted experiments to characterize a novel flexure mechanism to verify the performance of a theoretical structural augmentation, resulting in a publication in preparation.
- Collaborated with a team to design a flexure-based XY nanopositioning system with a focus on manufacturability, and carried the project through design, manufacturing, assembly, and open-loop testing, resulting in a patent and publication in preparation.
- Carried out thorough static and dynamic finite element analysis (FEA) to optimize flexure parameters for actuation stiffnesses, bearing stiffnesses, and resonant frequencies.
- Developed a framework with a graduate student to simulate and subsequently select the appropriate precision actuators and sensors based on flexure parameters and motion profile.

U-M BATTERY LAB

Research Assistant | University of Michigan

Sep 2021 - Jan 2022

Ann Arbor, MI

- Investigated the effect of fast formation on battery lifetime and electrochemistry by interpreting data from tests such as hybrid pulse power characterization (HPPC) and constant discharge.
- Analyzed diagnostic data from end-of-life lithium-ion pouch cells in Python to elucidate the effects of discharge temperature and presented the results in a digestible format.

MICROFLUIDICS AND SOFT MATTER GROUP

Research Assistant | The University of Hong Kong

May 2018 - Sep 2018

Hong Kong

- Conducted individual investigation on the structural strength of Aqueous Two Phase System (ATPS) capsules created with microfluidics for the application of timed medication release, triggered by capsule rupture from high voltage electric fields and Couette flow.
- Collaborated with graduate students on investigating the application of microfluidics in ATPS.

WORK EXPERIENCE

STEALTH-MODE ROBOTICS STARTUP

Jul 2023 - Present

Design Engineer

Ann Arbor, MI

- Led the development of a robust mechatronic system for a robotic device, successfully demonstrating the function and novelty of our product to prospective users.
- Leading the development of miniature manipulators for a robotic device, meeting crucial company milestones in a fast-paced environment with stringent deadlines.
- Managed projects for multiple initiatives, entailing responsibilities such as risk management, mechanical design, mechanism analysis, testing and validation, design for manufacturing, and coordination of manufacturing logistics, while ensuring adherence to industry standards.

PHOENIX CAPITAL

Jun 2021 - Aug 2021

Quantitative Analyst Intern

Hong Kong

- Investigated and implemented experimental trading algorithms based on buy-and-sell-point prediction from the perspective of survival statistical analysis.
- Devised profitable trading algorithms and technical indicators based on momentum trading.
- Wrote multiple internal tools and scripts in Excel VBA to streamline workflow in the office.

PUBLICATIONS

1. M. Radgolchin, D. Radtke, S. Rath, **R. Ho**, and S. Awtar, "Experimental characterization of a sandwich double parallelogram flexure mechanism," *Precision Engineering*, vol. 99, pp. 643–657, May 2026, doi: 10.1016/j.precisioneng.2026.02.002.

INVENTIONS

1. S. Awtar, **R. Ho**, and S. Rath, "Method of manufacture and assembly of XY flexure mechanism assembly," U.S. Patent 12,258,263 B1, Mar. 25, 2025.

HONORS

JAMES B. ANGELL SCHOLAR

Mar 2021 - Apr 2023

Awarded to students who achieve all A's (A+, A, A-) for two or more consecutive terms.

UNIVERSITY HONORS/DEAN'S LIST

Dec 2019 - Dec 2022

Awarded to students who earn a GPA of 3.5 or above during a term.

INDIVIDUAL PROJECTS

LASER SCANNING CONFOCAL MICROSCOPE

2022-2024

- Designed, simulated, and built a 3D printed XYZ flexure motion system capable of submicron (<200 nm) motion resolution over +/- 2.5mm motion range in all axes.
- Modified off-the-shelf open-loop stepper motors for closed-loop control to enable high-speed motion with minimal error (estimated <1 um) using step-loss compensation.
- Designed a confocal laser optical system with micron-scale optical resolution along with a gain-adjustable amplifier circuit to scan objects of varying albedos without saturation.
- Developed a library in C++ to control the microscope and process data from scans.

ULTRASONIC ACOUSTIC TWEEZERS

2022-2023

- Designed and built an enclosure that enables fine adjustment to the spacing between two opposing ultrasonic transducer arrays.
- Wrote custom Verilog HDL code for Altera Cyclone II FPGA to command phases of 50 ultrasonic transducers independently at $\pi/64$ resolution while concurrently receiving phase command input from microcontroller over SPI.
- Designed custom dual-layer PCB in KiCAD and hand-soldered SMD components.
- Developed a C++ library for real-time computation of the individual phases of 50 ultrasonic transducers for levitation, successfully suspending a small styrofoam ball at the focal point.

SOFTWARE EXPERIENCE

- **Programming:** MATLAB (Simulink, Simscape), LabView, Python, Java, Javascript, SQL, C++, C#, C, Verilog, Quartus II (ModelSim)
- **Mechanical:** Solidworks (CSWA Certified), ANSYS (Mechanical, Discovery), MSC Adams
- **Electronics:** LTSpice, KiCAD

SKILLS

- **Fabrication:** 3D Printing (SLA/FDM), Machining (Mill, Lathe), Metrology
- **Electronics:** Microcontroller and FPGA programming, Circuit design
- **Instrumentation:** LabVIEW (myRIO), Capacitance probes (Lion Precision), Optical linear encoders (Heidenhain, Renishaw), dSpace, Stereo/compound microscopy